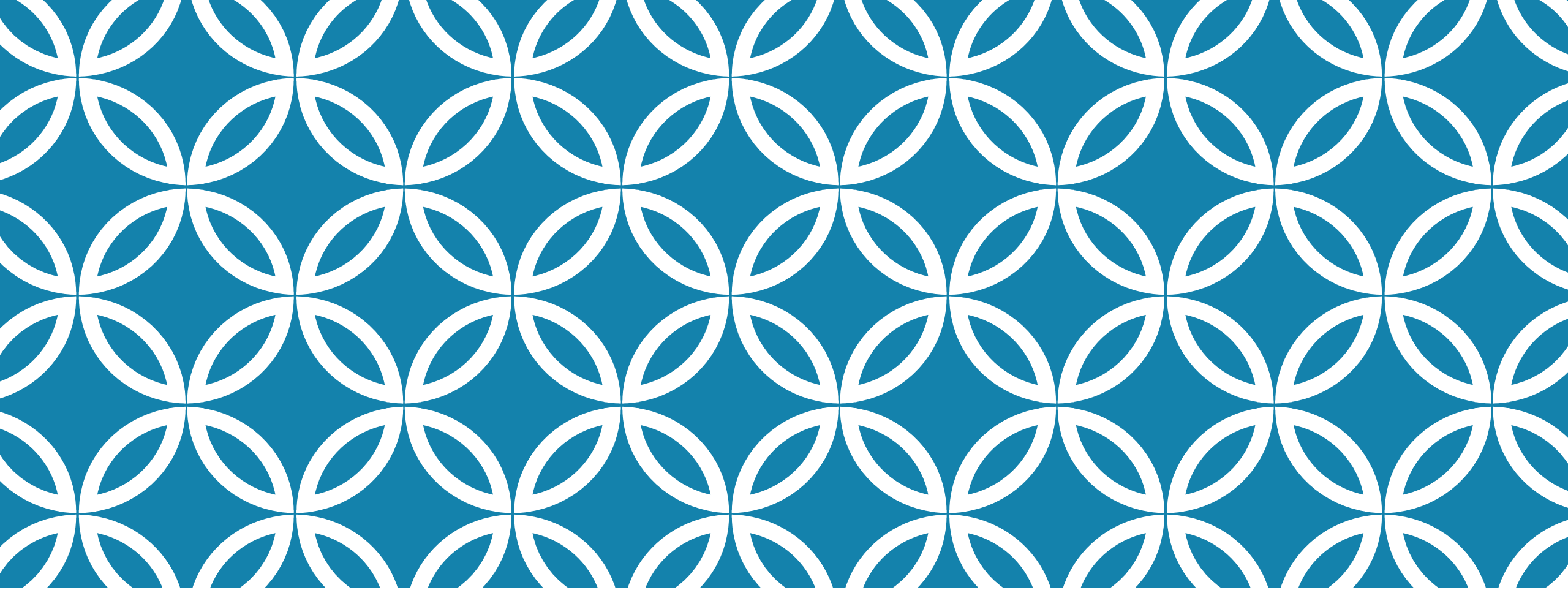
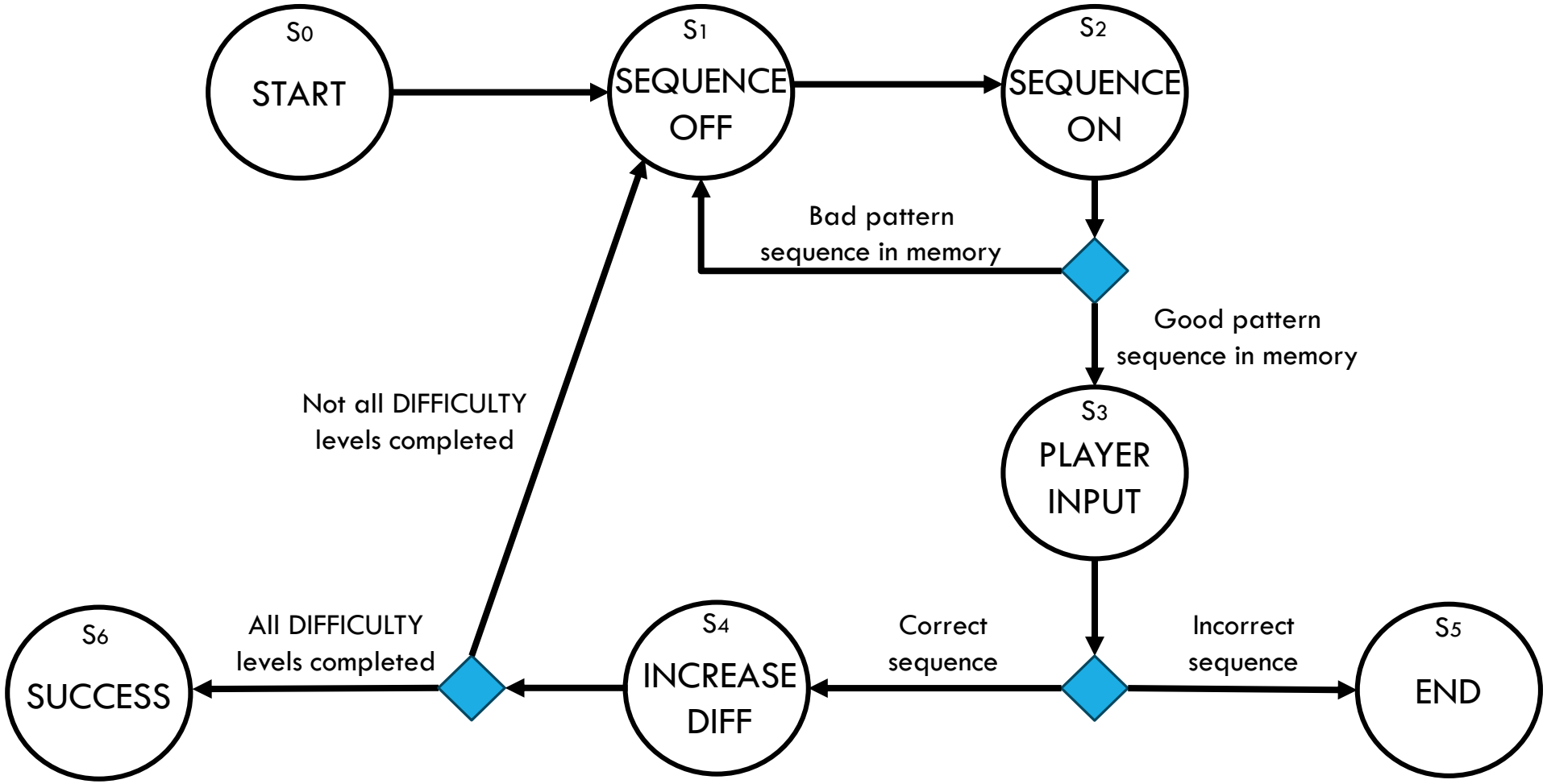


ENG GEN 225 DIGITAL CIRCUITS LAB 6

7-STATE STATE MACHINE



FINAL PROGRAMMING CHALLENGE

- Create the 7-State State Machine
- Synthesize your Simon Says project
- Load design to FPGA
- Show me that your project works

FINAL PROGRAMMING CHALLENGE

- Programming Challenge 1: Light up LED with 2, 3 and 4 buttons
  - Code (3 Projects)
  - Logic Flow Diagrams for 2-button case
- Programming Challenge 2: Using LEDs have FPGA count 0-15 in binary
  - Code
- Simulation Exercise
  - Truth Table for 2-button case
  - Testbench for 2-button case
  - Waveform Screenshot of 2-button case
- Programming Challenge 3: Testbench
  - Testbench for Programming Challenge 2
- Programming Challenge 4: Badge Door Opener
  - Code
  - Testbench
- Programming Challenge 5: 7-Segment LCD
  - Code
  - Testbench
  - Truth Table
- Final Programming Challenge: Simon Says
  - Code with all sub-modules